

A counterexample to Kourovka Notebook Problem 16.45

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Abstract

Let $b(G)$ be the largest size of an inclusion-minimal base, over all permutation representations of a finite group G , with bases taken for the permutation image. Let $\mu'(G)$ be the largest size of an independent subset of G , not necessarily generating G . We construct an explicitly specified affine group of order 100920 for which

$$b(G) = 3 < 4 = \mu'(G).$$

The upper bound covers faithful and nonfaithful actions and does not depend on enumerating the subgroups of the affine group. Exact, independently implemented Python and C++ verifiers certify the small matrix-group inputs and the explicit witnesses. The obstruction is a central involution of a complement which is not normal in the ambient affine group.

1 Definitions, source audit, and the finite reformulation

A subset S of a finite group X is *independent* if $x \notin \langle S \setminus \{x\} \rangle$ for each $x \in S$; the empty subset is allowed. Write $\mu'(X)$ for its maximum possible size. For an action $\rho : X \rightarrow \text{Sym}(\Omega)$, a base has pointwise stabilizer $\ker \rho$ in X . Minimal means inclusion-minimal. Write $b(X)$ for the maximum size of a minimal base over all actions, and $b_f(X)$ for the corresponding maximum restricted to faithful actions. Even if Ω is infinite, a base contains a finite subbase: choose a point moved by each nonidentity element of the finite permutation image. Consequently every inclusion-minimal base is finite.

These conventions give $b(1) = b_f(1) = \mu'(1) = 0$. Cameron's 2010 exposition uses b_2 for our b and μ^* for our μ' ; his 2014 paper uses b_2 and μ' respectively [2, 3]. In both, nonfaithful actions are allowed. The September 2026 notebook states Problem 16.45 and its normal-bottom meet-semilattice reformulation without an attached solution [1]. The separate September update does not mention 16.45. A focused literature check located no earlier resolution; this is a statement about the sources checked, not an independent guarantee of priority.

Proposition 1 (Subgroup formulation). *For a finite group X , $b(X)$ is the largest t for which subgroups $L_1, \dots, L_t$ satisfy*

$$N = \bigcap_{i=1}^t L_i \triangleleft X, \quad \bigcap_{j \neq i} L_j > N \quad (1 \leq i \leq t).$$

An empty intersection is X . The same description with $N = 1$ gives $b_f(X)$, and

$$b(X) = \max_{N \triangleleft X} b_f(X/N). \tag{1}$$

Proof. The point stabilizers of a minimal base have precisely these properties, with N the action kernel. Conversely, let such subgroups be given and let X act on the disjoint union of the left coset spaces X/L_i . Its kernel is

$$\bigcap_i \text{Core}_X(L_i) = \text{Core}_X\left(\bigcap_i L_i\right) = \text{Core}_X(N) = N.$$

Choose the identity coset in each component. Their common stabilizer is N , and deleting any one enlarges it, so these points are a minimal base. This argument uses cores only to compute the kernel; it does not replace the irredundant family by a family of cores. Formula (1) follows by factoring an action through its kernel and, conversely, pulling back a faithful quotient action. The empty family handles the trivial group and trivial actions consistently. $\square$

Call a subgroup family *meet-irredundant* if every member is essential to its total intersection, without any normality requirement. Let $c_\cap(X)$ be its largest size.

Lemma 2. *For every finite group X , $c_\cap(X) = \mu'(X)$. In particular, $b(X) \leq \mu'(X)$.*

Proof. For a meet-irredundant family $L_1, \dots, L_t$, choose

$$x_i \in \left(\bigcap_{j \neq i} L_j\right) \setminus L_i.$$

Every x_j with $j \neq i$ lies in L_i , whereas x_i does not. Therefore the x_i are distinct and independent. Conversely, from an independent set $\{x_1, \dots, x_t\}$ take $L_i = \langle x_j : j \neq i \rangle$. The element x_i witnesses the essentiality of L_i . Neither direction asserts that the total intersection is normal. $\square$

Lemma 3 (Restriction). *If $L_1, \dots, L_t$ is meet-irredundant in X , then, for each i ,*

$$t - 1 \leq \mu'(L_i).$$

If its total intersection is 1, the stronger bound $t - 1 \leq b_f(L_i)$ holds.

Proof. Inside L_i , the family $L_j \cap L_i$ for $j \neq i$ remains meet-irredundant: the original witness for L_j lies in L_i and witnesses this assertion. Its total intersection equals the original total intersection. Apply Lemma 2, or the faithful subgroup formulation when that intersection is 1. $\square$

For completeness, an irredundant family gives a top-preserving Boolean meet-semilattice embedding

$$A \mapsto \bigcap_{i \notin A} L_i \quad (A \subseteq \{1, \dots, t\}).$$

The witness x_i distinguishes images of sets differing at i ; intersection corresponds to intersection of index sets. The bottom is $\bigcap_i L_i$. There is no assertion that joins are preserved, or that this bottom is the identity. Thus the proof retains the precise semilattice convention in the notebook.

2 The complement and its two different base invariants

All matrices in the construction below are over $\mathbf{F}_{29}$. Put

$$A = \begin{pmatrix} 0 & 28 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 7 \\ 12 & 28 \end{pmatrix}, \quad H = \langle A, B \rangle \leq \mathrm{SL}_2(29), \quad Z = \{I, -I\}.$$

Direct multiplication gives

$$A^2 = B^3 = (AB)^5 = -I. \quad (2)$$

The standard spherical triangle presentation

$$A_5 = \langle a, b \mid a^2 = b^3 = (ab)^5 = 1 \rangle$$

and simplicity of A_5 show that $H/Z \cong A_5$: the projective image is a nontrivial quotient of this presentation, since $A \notin Z$. Consequently $|H| = 120$. The group is isomorphic to $\mathrm{SL}_2(5)$.

There is also a wholly finite certification of these facts, independent of the triangle presentation. The supplied verifiers generate exactly 120 matrices and construct an explicit isomorphism from $\mathrm{SL}_2(5)$ by mapping

$$\begin{pmatrix} 0 & 4 \\ 1 & 0 \end{pmatrix} \mapsto A, \quad \begin{pmatrix} 0 & 4 \\ 1 & 1 \end{pmatrix} \mapsto B.$$

The Python verifier additionally constructs the quotient map onto A_5 by mapping A to $(1\,2)(3\,4)$ and B to $(1\,3\,5)$, and checks that its kernel is exactly Z . Thus the small-group identifications can alternatively be read as explicit finite certificates, not as catalogue lookups.

We record the elementary group facts needed below. Every proper subgroup of A_5 lies in a subgroup isomorphic to A_4 , D_{10} , or S_3 . Here D_{10} denotes a group of order 10. One way to see this is to use the natural action on five letters. An intransitive subgroup either fixes a letter or has orbit sizes $2 + 3$. A proper transitive subgroup has order 5, 10, 15, 20, or 30. Orders 15 and 20 would give a normal Sylow 5-subgroup, whose normalizer in A_5 has order 10. Order 30 is impossible: an index-two quotient is trivial on every 3-cycle, and 3-cycles generate A_5 . The remaining cases lie in a Sylow-5 normalizer. This normalizer has order 10 because the order-20 normalizer in S_5 contains odd permutations.

Each of A_4, D_{10}, S_3 has independence number 2. For A_4 , a generating set contains a 3-cycle and some element outside its cyclic subgroup; these two generate A_4 , since there is no subgroup of order 6. Its proper subgroups have independence number at most 2. For a dihedral group of order $2q$, q prime, a generating set contains either a nontrivial rotation and a reflection, or two distinct reflections, and two such elements suffice. It follows, by deleting one element from any independent set in A_5 , that

$$\mu'(A_5) \leq 3. \quad (3)$$

The natural degree-five action has a minimal base of size 3, so equality holds. Simplicity, used above and below, also has an elementary check: the nonidentity conjugacy-class sizes are 15, 20, 12, 12, and no proper nontrivial union of classes together with the identity has order dividing 60.

Lemma 4. *The complement satisfies*

$$\mu'(H) = b(H) = 3, \quad b_f(H) = 2.$$

Proof. The only involution in $\mathrm{SL}_2(29)$ is $-I$: a matrix squaring to I is diagonalizable with eigenvalues ± 1 , and determinant one forces both signs to agree. Therefore every even-order subgroup of H

contains Z . Every odd-order subgroup injects into A_5 and, by the preceding subgroup description, is trivial or cyclic of order 3 or 5.

Let S be independent in H . If its images in H/Z , as an indexed family, are independent, then $|S| \leq 3$. Otherwise some $s \in S$ lies in KZ , where $K = \langle S \setminus \{s\} \rangle$, but does not lie in K . Thus Z is not contained in K , so K has odd order and is trivial or cyclic of prime order. Since $S \setminus \{s\}$ is independent, $|S| \leq 2$. This proves $\mu'(H) \leq 3$.

For the faithful invariant, a subgroup family with intersection 1 must have an odd-order member, since all even-order members contain Z . If this member is trivial, a minimal such family has size 1. If it has prime order, some other member intersects it trivially, so a minimal family has size at most 2. Nontrivial subgroups of orders 3 and 5 give a faithful family of size 2, proving $b_f(H) = 2$.

For explicit witnesses to the remaining lower bounds, let

$$R_1 = \begin{pmatrix} 0 & 28 \\ 1 & 0 \end{pmatrix}, \quad R_2 = \begin{pmatrix} 12 & 24 \\ 0 & 17 \end{pmatrix}, \quad R_3 = \begin{pmatrix} 25 & 3 \\ 4 & 4 \end{pmatrix}. \quad (4)$$

They belong to H , since $R_1 = A$, $R_2 = BAB^2$ and $R_3 = AR_2(AB)^2$. Direct multiplication gives $R_i^2 = -I$ and

$$|\langle R_2, R_3 \rangle| = 8, \quad |\langle R_1, R_3 \rangle| = 12, \quad |\langle R_1, R_2 \rangle| = 20.$$

For example, the corresponding products have orders 4, 6, 10, so their projective images generate dihedral groups of orders 4, 6, 10; adjoining the common central involution doubles these orders. The subgroup generated by all three has order divisible by $\text{lcm}(8, 12, 20) = 120$, and is therefore H . Hence the three matrices are independent and $\mu'(H) = 3$. Their three omission subgroups intersect in Z , and each is essential. Indeed, the first two intersect in $\langle R_3 \rangle$, of order 4, and the third cannot contain R_3 . Thus they also give $b(H) \geq 3$. The upper bound $b(H) \leq \mu'(H)$ finishes the proof. $\square$

3 A small affine lemma

Lemma 5. *Let p be an odd prime and let C be a cyclic 2-subgroup of $\mathbf{F}_p^\times$. For the faithful scalar semidirect product $L = \mathbf{F}_p \rtimes C$, $\mu'(L) \leq 2$.*

Proof. Put $P = (\mathbf{F}_p, +)$. A subgroup J with $J \cap P = 1$ injects into C , so is a cyclic 2-group and has independence number at most 1. Otherwise J contains P , and is $P \rtimes D$ for some $D \leq C$. The case $D = 1$ is cyclic of prime order. In the other cases, every generating set for J contains an element x whose projection generates D , because a cyclic 2-group has a unique maximal subgroup. If $x = (a, \lambda)$ and $|D| = m$, then

$$x^m = \left(a \sum_{j=0}^{m-1} \lambda^j, 1 \right) = 1,$$

so $\langle x \rangle$ is a complement of order m . Some other generator y is outside this complement. The subgroup $\langle x, y \rangle$ projects onto D ; if its intersection with P were trivial it would have order m and would equal $\langle x \rangle$, a contradiction. Thus it contains P and equals J . Every independent generating set of every subgroup has size at most 2. $\square$

Lemma 6. *Every line stabilizer in $H \leq \text{SL}_2(29)$ is cyclic of order dividing 4. Consequently H acts irreducibly on $V = \mathbf{F}_{29}^2$.*

Proof. For a line W , the scalar action on W defines a homomorphism $H_W \rightarrow \mathbf{F}_{29}^\times$. In a basis adapted to W , an element in its kernel has the form $\begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix}$, because its determinant is one. If $u \neq 0$, this element has order 29, which cannot divide $|H| = 120$. Thus the homomorphism is injective, and

$$|H_W| \mid \gcd(120, 28) = 4.$$

Since $|H| > 4$, the whole group cannot stabilize a line. $\square$

4 The affine counterexample and the all-actions upper bound

Set

$$G = V \rtimes H, \quad V = \mathbf{F}_{29}^2,$$

with multiplication

$$(v, h)(w, k) = (v + hw, hk).$$

The group order is $29^2 \cdot 120 = 100920$; write $\pi : G \rightarrow H$ for projection. This description specifies the action, not just the abstract factors of a semidirect product.

Lemma 7. *If $K \leq G$ does not contain V , then $b_f(K) \leq 2$. More precisely, if $K \cap V = 1$, then $\mu'(K) \leq 3$ and $b_f(K) \leq 2$; if $\dim_{\mathbf{F}_{29}}(K \cap V) = 1$, then $\mu'(K) \leq 2$.*

Proof. Every additive subgroup of V is an $\mathbf{F}_{29}$ -subspace, since 29 is prime. If $K \cap V = 1$, then π embeds K in H . Both μ' and b_f are monotone for subgroups: for the latter, use the same subgroup family with trivial intersection in the larger group. Apply Lemma 4.

Otherwise $W = K \cap V$ is a line, normal in K , so $\pi(K) \leq H_W$ is cyclic of order dividing 4. A Sylow 2-subgroup of K maps isomorphically onto $\pi(K)$ and complements W . Its action on W is faithful by Lemma 6. Thus K is isomorphic to a group covered by Lemma 5, which gives $\mu'(K) \leq 2$ and hence $b_f(K) \leq 2$. These are all possibilities for a subgroup not containing V . $\square$

Proposition 8. $b_f(G) \leq 3$.

Proof. Let $K_1, \dots, K_t$ be any meet-irredundant family with total intersection 1. Some K_i does not contain V , since otherwise the intersection would contain V . Lemmas 3 and 7 give

$$t - 1 \leq b_f(K_i) \leq 2.$$

No transitivity, maximality, or conjugacy assumption on the subgroups was made. $\square$

Proposition 9. $b(G) \leq 3$, including all nonfaithful actions.

Proof. Every nontrivial normal subgroup $N \triangleleft G$ contains V . To prove this, irreducibility gives $N \cap V = 1$ or V . In the first case $[N, V] \leq N \cap V = 1$, so $N \leq C_G(V)$. The matrix action is faithful, hence $C_G(V) = V$, forcing $N = 1$. This contradicts the choice of N .

It follows that every nonfaithful action of G factors through $G/V = H$, and its minimal bases have size at most $b(H) = 3$. Faithful actions are covered by Proposition 8. This proves the stated bound for every representation. $\square$

5 Explicit lower bounds and the exact independence number

Let $e_1 = (1, 0)^\top$ and $e_2 = (0, 1)^\top$. With the matrices in (4), define

$$S = \{(e_1, I), (0, R_1), (0, R_2), (0, R_3)\} \subseteq G.$$

The three linear elements project to the independent triple in H . Thus no one of them is generated by the other three elements of S , since its projection would then be generated by the other two matrices. The translation (e_1, I) is not in the zero-translation complement. Therefore S is independent and $\mu'(G) \geq 4$.

In fact S generates G : it contains generators for H , and the orbit of a nonzero vector under an irreducible group spans V . Its omission subgroups have the following explicitly verified orders:

Omitted element	Order of the subgroup generated by the others
(e_1, I)	120
$(0, R_1)$	$841 \cdot 8 = 6728$
$(0, R_2)$	$841 \cdot 12 = 10092$
$(0, R_3)$	$841 \cdot 20 = 16820$

The latter three formulas also follow structurally: each corresponding pair subgroup of H has order greater than 4, so cannot stabilize a line and is irreducible on V .

Proposition 10. $\mu'(G) = 4$.

Proof. It remains to give the upper bound. By Lemma 2, consider any meet-irredundant family $K_1, \dots, K_t$, with no assumption on its total intersection. If all members contain V , quotienting by the genuinely normal subgroup V gives an irredundant family in H , so $t \leq 3$. Otherwise choose $K_i \not\supseteq V$. If $K_i \cap V = 1$, then $t - 1 \leq \mu'(K_i) \leq 3$. If $K_i \cap V$ is a line, then $t - 1 \leq \mu'(K_i) \leq 2$. In either case $t \leq 4$. $\square$

There is also a faithful minimal base of size 3. Define subgroups

$$\begin{aligned} L_1 &= \langle (e_1, I), (0, -I) \rangle, \\ L_2 &= \langle (e_2, I), (0, -I) \rangle, \\ L_3 &= \langle (e_1 + e_2, I), (2e_1, -I) \rangle. \end{aligned}$$

Each has order 58. Their pairwise intersections are

$$\begin{aligned} L_1 \cap L_2 &= \{1, (0, -I)\}, \\ L_1 \cap L_3 &= \{1, (2e_1, -I)\}, \\ L_2 \cap L_3 &= \{1, (-2e_2, -I)\}. \end{aligned}$$

Their total intersection is 1. The three identity cosets in the disjoint union $G/L_1 \sqcup G/L_2 \sqcup G/L_3$ therefore form a faithful minimal base of size 3. This particular action has degree $3(100920/58) = 5220$.

Theorem 11 (Counterexample). *For the explicitly specified group*

$$G = \mathbf{F}_{29}^2 \rtimes \left\langle \begin{pmatrix} 0 & 28 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 2 & 7 \\ 12 & 28 \end{pmatrix} \right\rangle,$$

one has

$$|G| = 100920, \quad b_f(G) = b(G) = 3 < 4 = \mu'(G).$$

Consequently Kourouka Notebook Problem 16.45 has a negative answer under the stated all-representations conventions.

6 The normality obstruction is explicit

Let $M_i = \langle S \setminus \{s_i\} \rangle$ for the four elements of S in the order displayed above. Then

$$\bigcap_{i=1}^4 M_i = Z_0, \quad Z_0 = \{(0, I), (0, -I)\}.$$

Indeed, $M_1 = H$ and the other three are the preimages of the three pair subgroups in Lemma 4, whose intersection is Z . The subgroup Z_0 is not normal in G , because

$$(e_1, I)(0, -I)(-e_1, I) = (2e_1, -I) \notin Z_0.$$

Thus this independent 4-set gives an irredundant intersection family, and a Boolean meet-semilattice of rank 4, with nonnormal bottom. The theorem proves that no rank-four embedding can have normal bottom.

Taking normal cores does not repair this example. Simplicity of $H/Z \cong A_5$ implies that the three proper pair subgroups of H have core Z . Since M_2, M_3, M_4 contain V , their cores in G are $V \rtimes Z$. The complement $M_1 = H$ has trivial core; for example, its intersection with its conjugate by (e_1, I) is trivial. Hence the four cores are

$$1, \quad V \rtimes Z, \quad V \rtimes Z, \quad V \rtimes Z.$$

The first member alone gives their total intersection. All four-way essentiality has been lost. No quotient by the nonnormal subgroup Z_0 is used anywhere in the argument.

7 Exact certification and adversarial tests

The final verifiers are standalone and use exact integers only. Python uses only its standard library. The second implementation is C++17 and reads no Python-produced group table. Both reconstruct the complement and enumerate *all 76 actual subgroups*, not just conjugacy representatives. Their subgroup sets were compared as sets of actual matrices and agree.

Coverage of the subgroup enumeration follows by adjoining every element outside every subgroup already found, starting with the identity subgroup. For any subgroup K , a generating sequence for K is a path in this search. The two verifiers check all $\binom{75}{4} = 1215450$ four-element families of proper subgroups of H and find none meet-irredundant. Every subfamily of an irredundant family is irredundant, so this excludes all larger ranks as well. They then check all $\binom{75}{3} = 67525$ three-element families and find none both faithful and minimal. Together with the four-family check, this independently certifies $\mu'(H) = 3$ and $b_f(H) = 2$ using the explicit lower witnesses. The mathematical proofs above do not require trusting an optimizer.

They also verify the independent affine 4-set, generate all 100920 elements of G without storing its full multiplication table, check the omission orders and the nonnormal intersection, and verify the faithful 3-family. The Python verifier explicitly checks all 30 line stabilizers, all of which are in fact cyclic of order 4, and includes the following adversarial test.

The characteristic restriction cannot be discarded. Over $\mathbf{F}_{11}$, use

$$A_{11} = \begin{pmatrix} 0 & 10 \\ 1 & 0 \end{pmatrix}, \quad B_{11} = \begin{pmatrix} 0 & 2 \\ 5 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 6 & 10 \\ 0 & 2 \end{pmatrix}.$$

The first two generate a complement of order 120; C has order 10 and stabilizes the distinct lines $W_1 = \langle (1, 0) \rangle$ and $W_2 = \langle (1, 4) \rangle$. In $\mathbf{F}_{11}^2 \rtimes \langle A_{11}, B_{11} \rangle$, the four subgroups

$$W_1 \rtimes \langle C \rangle, \quad W_2 \rtimes \langle C \rangle, \quad V \rtimes \langle C^5 \rangle, \quad V \rtimes \langle C^2 \rangle$$

have total intersection 1. Their deletion intersections have respective orders 11, 11, 5, 2, so they give a faithful minimal base of size 4. This explicitly falsifies the tempting but incorrect claim that every such characteristic gives the counterexample. The equality $\gcd(120, 28) = 4$, and the resulting line-subgroup bound, are essential.

Earlier exploratory runs used integer-programming models on complete small subgroup lattices. Their code and logs are retained for provenance, but no floating-point MILP upper bound is a premise of this proof. The exact certificate and the structural all-actions argument replace those exploratory results. No Lean proof has been compiled, and no external peer review or publication is claimed.

References

- [1] E. I. Khukhro and V. D. Mazurov (eds.), *Unsolved Problems in Group Theory: The Kourovka Notebook*, 21st edition, September 2026 update, Problem 16.45, printed p. 100. <https://kourovkanotebookorg.wordpress.com/wp-content/uploads/2026/09/21tkr.pdf>. Separate update: <https://kourovkanotebookorg.wordpress.com/wp-content/uploads/2026/09/21upd.pdf>.
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